Foundations of Programming

Assignment No.: 19

PROBLEM STATEMENT: Create Structure EMPLOYEE for storing details (Name, Designation, gender, Date of Joining and Salary). Define function members to compute a) total number of employees in an organization b) count of male and female employee c) Employee with salary more than 10,000 d) Employee with designation “Asst Manager”

OBJECTIVE:

- To understand the use of structures in C programming.
- To store and manage employee records using structures.
- To perform operations such as counting and filtering employee data

THEORY:

In the C programming language, a structure is a user-defined data type that groups related variables of different data types under a single name. Structures are widely used for representing records such as employee information. In this program, a structure named EMPLOYEE is created to store employee details such as name, designation, gender, date of joining, and salary. By using loops and conditional statements, the program can calculate the total number of employees, count male and female employees, find employees whose salary is greater than 10,000, and display employees having the designation 'Asst Manager'.

PLATFORM: Linux

```
#include <stdio.h>

#include <string.h>

struct Employee

{

   char name[50];

   char designation[50];

   char gender[10];
```

```
    char doj[20];
    float salary;
};
int main()
{
    struct Employee emp[100];
    int n,i,male=0,female=0;
    printf("Enter number of employees: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Name: ");
        scanf("%s",emp[i].name);
        printf("Designation: ");
        scanf("%s",emp[i].designation);
        printf("Gender: ");
        scanf("%s",emp[i].gender);
        printf("Date of Joining: ");
        scanf("%s",emp[i].doj);
        printf("Salary: ");
        scanf("%f",&emp[i].salary);
    for(i=0;i<n;i++)
    {
```

```
        if(strcmp(emp[i].gender,"Male")==0)
            male++;
        else
            female++;
    }
    printf("Total Employees = %d\n",n);
    printf("Male Employees = %d\n",male);
    printf("Female Employees = %d\n",female);
    printf("Employees with salary more than 10000:\n");
    for(i=0;i<n;i++)
        if(emp[i].salary>10000)
            printf("%s\n",emp[i].name);
    printf("Employees with designation AsstManager:\n");
    for(i=0;i<n;i++)
        if(strcmp(emp[i].designation,"AsstManager")==0)
            printf("%s\n",emp[i].name);
    return 0;
}
```

INPUT:

```c
#include <stdio.h>
#include <string.h>
struct Employee
{
    char name[50];
    char designation[50];
    char gender[10];
    char doj[20];
    float salary;
};
int main()
{
    struct Employee emp[100];
    int n,i,male=0,female=0;
    printf("Enter number of employees: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Name: ");
        scanf("%s",emp[i].name);
        printf("Designation: ");
        scanf("%s",emp[i].designation);
        printf("Gender: ");
        scanf("%s",emp[i].gender);
        printf("Date of Joining: ");
        scanf("%s",emp[i].doj);
        printf("Salary: ");
        scanf("%f",&emp[i].salary);

    for(i=0;i<n;i++)
    {
        if(strcmp(emp[i].gender,"Male")==0)
            male++;
        else
            female++;
    }
    printf("Total Employees = %d\n",n);
    printf("Male Employees = %d\n",male);
    printf("Female Employees = %d\n",female);
    printf("Employees with salary more than 10000:\n");
    for(i=0;i<n;i++)
        if(emp[i].salary>10000)
            printf("%s\n",emp[i].name);
    printf("Employees with designation AsstManager:\n");
    for(i=0;i<n;i++)
        if(strcmp(emp[i].designation,"AsstManager")==0)
            printf("%s\n",emp[i].name);
```

Employee details such as Name, Designation, Gender, Date of Joining and Salary.

OUTPUT: Displays total number of employees, count of male and female employees, employees with salary greater than 10,000, and employees with designation 'Asst Manager'.

```
Enter number of employees: 2
Name: Anuradha
Designation: Salesperson
Gender: Female
Date of Joining: may
Salary: 50000
Name: Vicky
Designation: Manager
Gender: male
Date of Joining: august
Salary: 6000000
Total Employees = 2
Male Employees = 0
Female Employees = 2
Employees with salary more than 10000:
Anuradha
Vicky
Employees with designation AsstManager:
```

SAMPLE INPUT:
Enter number of employees: 2
Employee 1: Ravi, AsstManager, Male, 12-05-2022, 15000
Employee 2: Anita, Clerk, Female, 10-01-2023, 8000
SAMPLE OUTPUT:

Total number of employees = 2
Male employees = 1
Female employees = 1
Employees with salary more than 10000: Ravi
Employees with designation AsstManager: Ravi

CONCLUSION: Thus, the C program successfully demonstrates the use of structures to store employee details and perform various operations such as counting employees, identifying employees with salary greater than 10,000, and displaying employees with the designation 'Asst Manager'.

FAQs:

1. How does a structure in the C programming language help in organizing multiple related data items such as employee details?

2. Why are structures preferred over simple variables or arrays when managing records like employee information?
3. How can conditional statements and functions be used with structures to filter employees based on salary or designation?